

# Power Supply Systems

## Part II Roadmap

### Phase 0: Absolute-Zero Orientation

**Aim:** To understand what this Part-II syllabus is actually about before solving any question.

**Style:** Simple English, SN ma'am style, diagram first, formula after understanding.

**Target:** Semester-question-ready preparation.

**Subject:** Power Supply Systems

**Section:** Part II

**Phase:** 0 – Foundation and Reverse Engineering

**Level:** Absolute zero to course map clarity

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## 1 What Phase 0 Means

### Phase 0 Goal

Phase 0 is not for heavy derivation. Phase 0 is for building the **map of the subject**. Before learning distributor numericals, Kelvin's law, fuse, earthing, or substation, we must know where each topic lives inside the power supply system.

### Definition

**Power Supply System** means the complete arrangement by which electrical power is generated, transmitted, distributed, protected, and finally supplied to consumers safely and economically.

In Part II, we mainly study what happens **after generation and major transmission**. We focus on:

Substation → Distribution → Distributor Design → Wiring → Earthing → Fuse and Protection

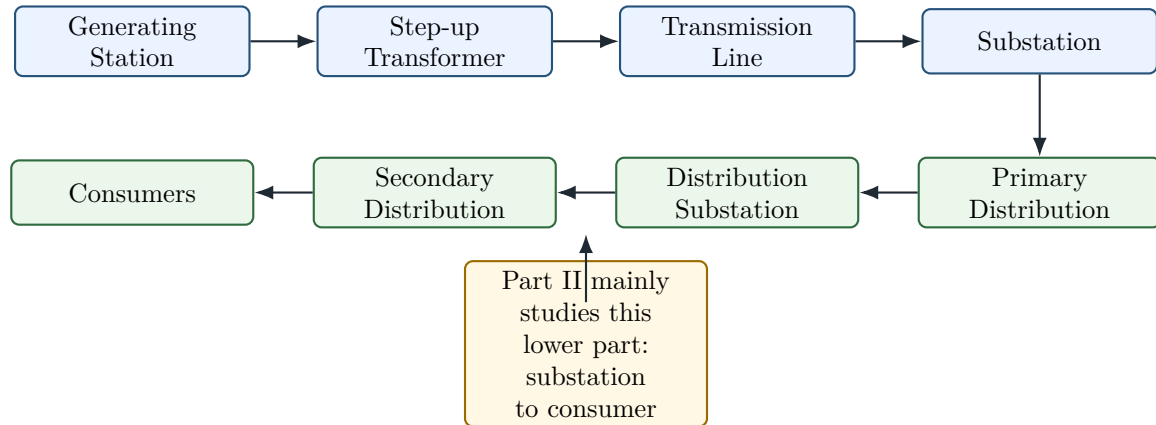
### Memory Hook

Think of Part II as the journey of power from the substation to the consumer's switchboard.

## 2 Complete Part-II Syllabus in Simple Words

| Syllabus Block                                 | Simple Meaning                                                                                                                                                     |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power distribution system                      | How power is distributed from substation to consumers. It includes primary distribution, secondary distribution, and comparison of different distribution systems. |
| Types of conductors in distribution system     | Which conductors are used in overhead and distribution lines, such as stranded conductor, ACSR conductor, copper, aluminium, etc.                                  |
| Comparison of distribution systems             | Comparison between radial, ring main, interconnected, AC/DC and other supply arrangements.                                                                         |
| Distributor design                             | How to calculate voltage drop, current in each section, minimum voltage point and load point voltage in a distributor.                                             |
| Radial and ring main                           | Different methods of feeding consumers. Radial system is simple but less reliable. Ring main is more reliable.                                                     |
| Current and voltage profiles along distributor | How current changes section by section and how voltage falls along the distributor.                                                                                |
| Economics of feeder design                     | How to choose economical conductor size. Kelvin's law belongs here.                                                                                                |
| Electrical wiring and installation             | Domestic, commercial and industrial wiring rules, main, sub-main and sub-circuit wiring.                                                                           |
| Earthing practice                              | How metallic parts are connected to earth for safety. Includes pipe earthing and plate earthing.                                                                   |
| Testing of installation                        | Basic checking before energising electrical installation.                                                                                                          |
| Special lighting connections                   | Lighting circuits and common practical wiring arrangements.                                                                                                        |
| Conductors, fuse and disconnecting devices     | Stranded conductor, ACSR, fuse, HRC fuse, isolator, circuit breaker and related protective devices.                                                                |

### 3 The Big Picture Diagram



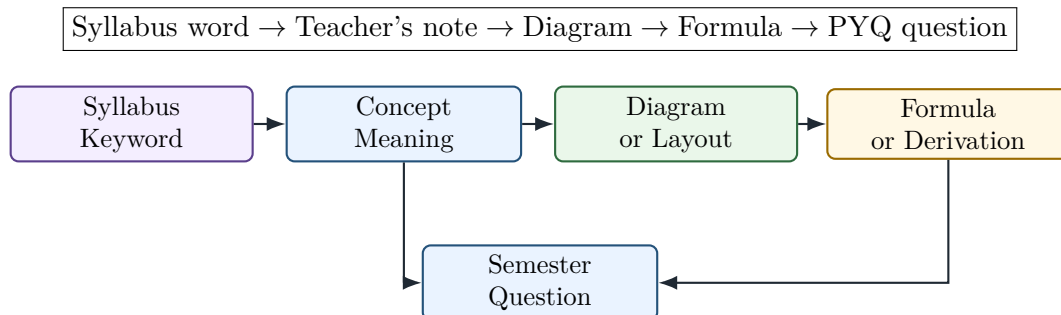
#### Semester Answer Style

In the exam, when asked about the structure of distribution system, draw a simple power-flow diagram first. Then explain each part in one or two lines.

### 4 Reverse Engineering: How Questions Are Formed

#### 4.1 Main Idea

Semester questions are not random. They are formed from repeated patterns.



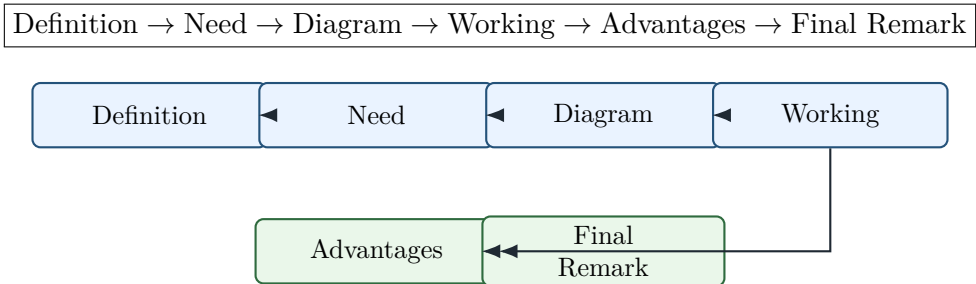
#### 4.2 Example of Reverse Engineering

| Syllabus Word              | Concept                                  | Possible Semester Question                                                     |
|----------------------------|------------------------------------------|--------------------------------------------------------------------------------|
| Distributor design         | Voltage drop and minimum potential point | Derive voltage drop expression for a distributor fed at one end or both ends.  |
| Ring main                  | Closed-loop distributor                  | Solve ring main distributor and find voltage at load points.                   |
| Economics of feeder design | Kelvin's law                             | State and prove Kelvin's law. Find economical conductor size.                  |
| Fuse                       | Protection device                        | Explain HRC fuse with diagram. Define pre-arcing time and prospective current. |
| Earthing practice          | Safety connection to earth               | Explain pipe and plate earthing. Mention factors affecting earth resistance.   |
| Busbar layout              | Substation bus arrangement               | Explain single busbar or double busbar arrangement with diagram.               |

## 5 How to Write Answers in SN Ma'am Style

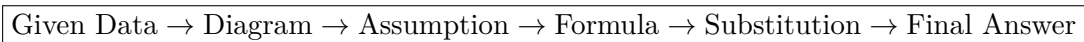
### 5.1 Theory Answer Format

For every theory question, use this pattern:



### 5.2 Numerical Answer Format

For every numerical, use this pattern:



#### Common Mistake

Do not start a numerical directly with calculation. First draw the line diagram and mark loads, currents, distances and voltage.

## 6 Basic Words You Must Know Before Phase 1

### 6.1 Feeder

#### Definition

A **feeder** is a conductor which carries power from substation to the distribution area. Normally, no tapping is taken from a feeder.

It is designed mainly from the point of view of **current carrying capacity**.

Feeder  $\Rightarrow$  current capacity is important

### 6.2 Distributor

#### Definition

A **distributor** is a conductor from which many tappings are taken for supplying consumers.

It is designed mainly from the point of view of **voltage drop**.

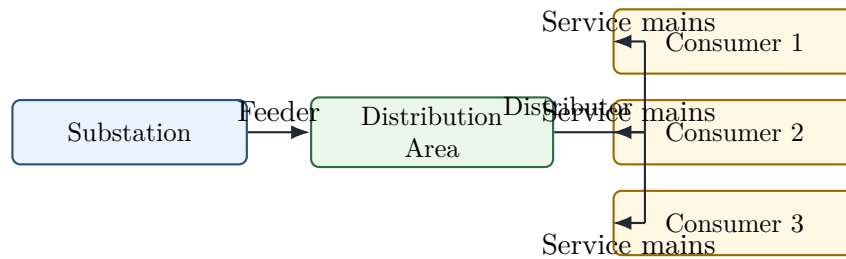
Distributor  $\Rightarrow$  voltage drop is important

### 6.3 Service Mains

#### Definition

**Service mains** are the conductors which connect consumer terminals to the distributor.

## 6.4 Feeder–Distributor–Service Mains Diagram



### Memory Hook

Feeder feeds the area. Distributor distributes to many consumers. Service mains serve one consumer.

## 7 Course Blocks in Best Learning Sequence

| Phase | Block                              | What You Will Learn                                                                          |
|-------|------------------------------------|----------------------------------------------------------------------------------------------|
| 0     | Orientation                        | Full map of Part-II, question pattern, answer-writing method.                                |
| 1     | Distribution Basics                | Primary/secondary distribution, feeder, distributor, service mains, radial and ring systems. |
| 2     | Conductor and Transmission Economy | Types of conductors, stranded conductor, ACSR, conductor-material comparison.                |
| 3     | Kelvin's Law                       | Economical conductor size, cost curve, derivation and numerical.                             |
| 4     | Distributor Fed at One End         | Current loading, voltage drop, minimum voltage at far end.                                   |
| 5     | Distributor Fed at Both Ends       | Minimum potential point, equal and unequal end voltage cases.                                |
| 6     | Ring Main and Inter-connector      | Ring distributor, KVL method, voltage at load points.                                        |
| 7     | Three-Wire DC and Unbalanced Loads | Neutral current, balancer, 3-phase 4-wire idea.                                              |
| 8     | Fuse and Protection                | Fuse, HRC fuse, relay, circuit breaker, isolator.                                            |
| 9     | Substation and Busbar              | Substation classification, equipment, single and double busbar.                              |
| 10    | Earthing and Wiring                | Pipe earthing, plate earthing, LT/MT/HT service connection, house wiring rules.              |

## 8 Why This Sequence Is Best

### 8.1 Reason 1: First Learn the Path of Power

Before solving voltage drop, you must know where voltage drops.

Substation → Feeder → Distributor → Service Mains → Consumer

### 8.2 Reason 2: Then Learn Economy

After knowing the path, we ask:

How thick should the conductor be?

This leads to:

Conductor comparison and Kelvin's law

### 8.3 Reason 3: Then Learn Distributor Numericals

After conductor economy, we ask:

What voltage will reach each consumer?

This leads to:

Voltage drop, current loading, minimum potential

### 8.4 Reason 4: Finally Learn Protection and Safety

After supplying power, we must protect people and equipment.

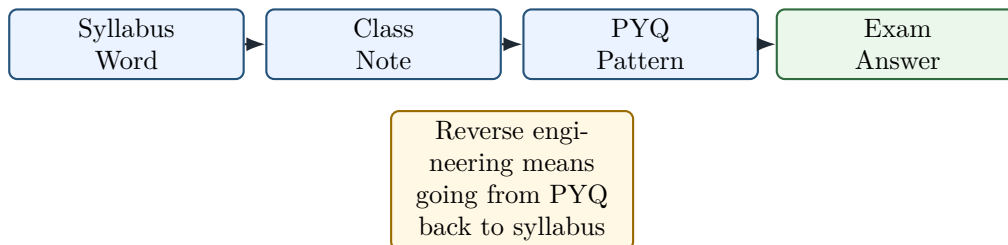
Fuse → Circuit breaker → Relay → Earthing

## 9 Core Diagrams to Remember from Phase 0

### 9.1 Diagram 1: Power Reaches Consumer



### 9.2 Diagram 2: How a Semester Question Is Built



## 10 Minimum Formula Awareness Before Starting

Phase 0 does not require derivation mastery. But you must recognise these formulas.

| Topic                    | Formula                         | Meaning                                 |
|--------------------------|---------------------------------|-----------------------------------------|
| Power in DC              | $P = VI$                        | Power equals voltage times current.     |
| Power in single-phase AC | $P = VI \cos \phi$              | Power factor affects real power.        |
| Power in three-phase AC  | $P = \sqrt{3}V_L I_L \cos \phi$ | Used in 3-phase systems.                |
| Copper loss              | $P_{\text{loss}} = I^2 R$       | Loss due to resistance.                 |
| Resistance               | $R = \rho l / a$                | Resistance decreases if area increases. |
| Voltage drop             | $\Delta V = IR$                 | Voltage lost in a conductor section.    |

|                   |                                   |                                    |
|-------------------|-----------------------------------|------------------------------------|
| Kelvin's law idea | Capital charge = Energy-loss cost | Economic conductor size condition. |
|-------------------|-----------------------------------|------------------------------------|

### Common Mistake

Do not memorise formulas without knowing the diagram. In PSS Part-II, formula comes from current path.

## 11 Predictable Semester Question Families

| Question Family              | Expected Question Form                                                               |
|------------------------------|--------------------------------------------------------------------------------------|
| Distribution basics          | Explain feeder, distributor and service mains. Compare radial and ring main systems. |
| Conductor comparison         | Compare volume of conductor material for two systems. State assumptions.             |
| Kelvin's law                 | State and prove Kelvin's law. Find economical conductor size.                        |
| Distributor fed at one end   | Find voltage at each load point and minimum voltage.                                 |
| Distributor fed at both ends | Find current supplied from both ends, point of minimum potential and voltage.        |
| Ring distributor             | Find current in each section and voltage at each load point.                         |
| Fuse                         | Define prospective current, pre-arcing time, HRC fuse. Draw and explain.             |
| Substation                   | Classify substations. Explain equipment, busbar, circuit breaker, isolator.          |
| Earthing                     | Explain pipe earthing, plate earthing, factors affecting earth resistance.           |
| Wiring                       | Mention general stipulations of service connection and house wiring.                 |



## 12 Phase 0 Final Memory Sheet

### One-Page Memory Summary

1. Part-II is mainly from **substation to consumer**.
2. Feeder is designed for **current carrying capacity**.
3. Distributor is designed for **voltage drop**.
4. Service mains connect **consumer to distributor**.
5. Kelvin's law gives **economic conductor size**.
6. Distributor numericals need **current in each section**.
7. Ring distributor is solved by **KVL around the loop**.
8. Fuse protects circuit by **melting during excess current**.
9. Earthing protects humans and equipment by giving **low resistance path to earth**.
10. Best exam answer style is:

Definition → Diagram → Formula → Stepwise Explanation

## 13 What You Must Be Able to Say After Phase 0

After Phase 0, you should be able to say:

*Power is generated at the generating station and transmitted at high voltage. Near the load centre, it reaches substations. From substations, feeders carry power to distribution areas. Distributors supply many consumers through service mains. In Part-II, we study how this distribution is designed, protected, earthed and economically planned.*

### Semester Answer Style

This paragraph is your absolute-zero oral answer. If you can say this clearly, you are ready to start Phase 1.

## End of Phase 0

### Next Phase: Phase 1 – Distribution Basics

Primary distribution, secondary distribution, feeder, distributor, service mains, radial system, ring main system.